

Learning Queue

Wednesday, July 1, 2026 2:15 PM

HIGH

0. 3D Priors
 1. Research the motivation of inclusion, what approaches have looked like, what current research is
 2. file:///C:/Users/sevmy/Downloads/Dissertation_Wei_Zhang-1.pdf#subsection.3.2.2
 3. Dynamic Visual SLAM using a General 3D Prior
1. LOOP CLOSURE
1. DUST3R/MASt3R/Fast3R
 - ^ Real time reconstruction of 3D spaces using an alternative approach from typical interest point/ descriptor based slam
1. SEMANTIC INCORPORATION TO MAPS
 1. Can you add this information to interest points or just create a voxel type map?
2. Mapping updates / training approaches that incorporate forgetting
3. INDOOR SPECIFIC APPROACHES
4. How fast are all of these approaches and what are the hardware requirements?
5. [Privacy](#)
6. WHAT HAVE ACCESSIBILITY ENABLING USES OF THIS LOOKED LIKE?
7. VPR vs other approaches
8. Semantic VPRs?
9. Continual learning VPR
10. What do demo's look like in this field?

Misc

Neural Implicit Fields (This is the broad category containing NeRF)

Youtube video: [Implicit Neural Representations: From Objects to 3D Scenes](#)

Cornel PDF: <https://www.cs.cornell.edu/courses/cs6670/2021fa/implicit.pdf>

Medium to Low

1. Manhattan type environment approaches (very applicable to hallways)
2. Loop Closing
 1. how is it done? What are modern methods for achieving this?
 2. file:///C:/Users/sevmy/Downloads/Dissertation_Wei_Zhang-1.pdf#subsection.3.3.1

QUESTIONS

- What is the role of COLMAP in SfM?
- Why is the accuracy of COLMAP important?
- What is the difference between SfM and SLAM?
- DUST4R is both?

Github list of key papers in the field uploaded 2 weeks ago

[End-to-End-3D-Reconstruction-Paper-List](#)

3D Reconstruction

Scalable

Self/Weak-Supervised

Semantic

Dynamic

Generation

Novel View Synthesis